

What is Entity Framework Core

- **ORM (Object-Relational Mapper)**
- **Lightweight, Extensible, and Open Source**
- **Works Cross Platform**
- **Microsoft's Official Data Access Platform**

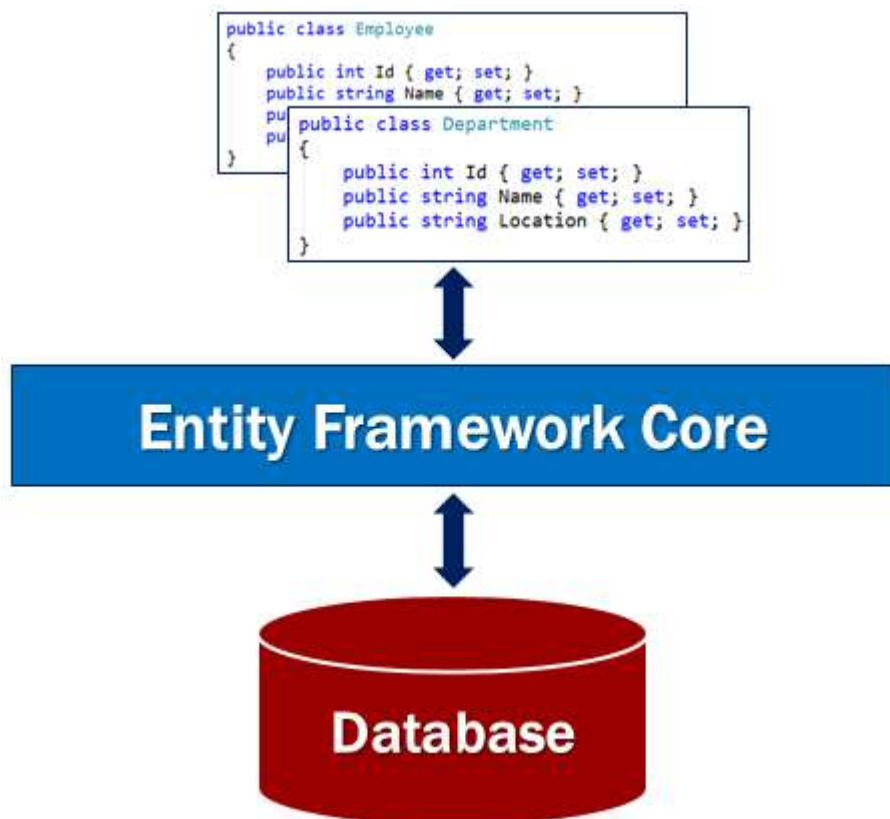
ORM stands for Object-Relational Mapper and it enables developers to work with a database using business objects. As a developer we work with the application business objects and the ORM generates the SQL that the underlying database understands. In-short, an ORM, eliminates the need for most of the data-access code that developers usually need to write.

Domain Classes

```
public class Employee
{
    public int Id { get; set; }
    public string Name { get; set; }
    public string Email { get; set; }
    public Department Department { get; set; }
}
```

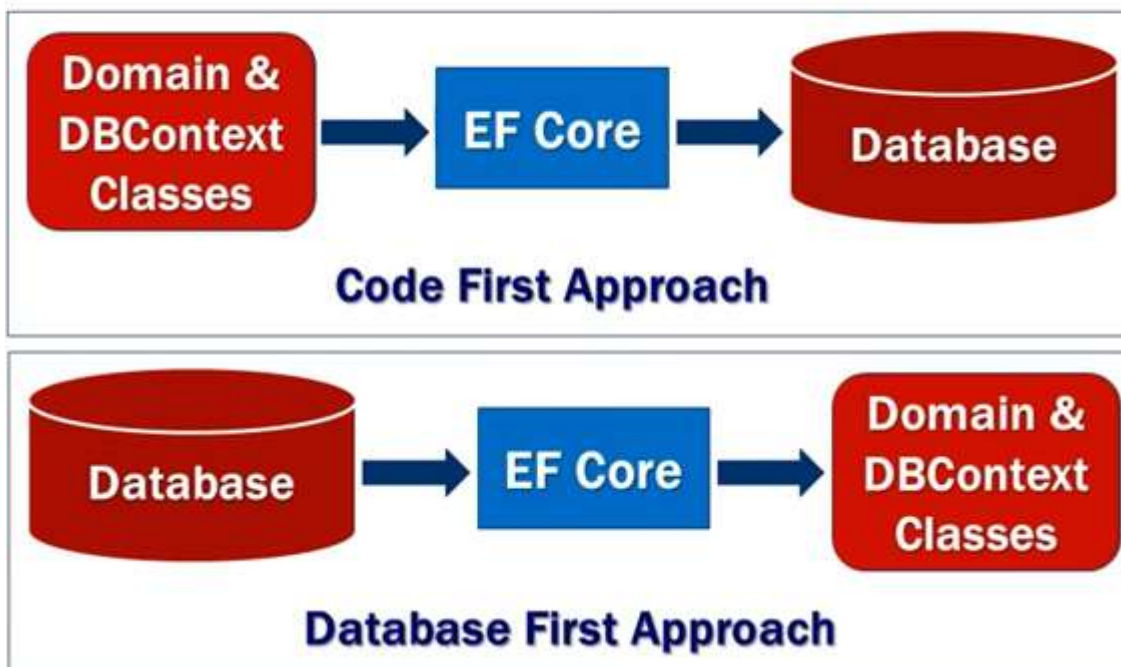
```
public class Department
{
    public int Id { get; set; }
    public string Name { get; set; }
    public string Location { get; set; }
}
```

ORM спестява много код за read, insert, update, delete DB tables.



Има два начина за връзка между класовете и ДБ: Първо код, после БД или обратното:

Code First & Database First Approach



Plug-in libraries = DB Providers (NuGet Packages): <https://docs.microsoft.com/en-us/ef/core/providers/?tabs=dotnet-core-cli> – съдържат цялата характерна функционалност, необходима за осъществяване на специфичните услуги.

	NuGet Package	Supported database engines	Maintainer / Vendor	Notes / Requirements	Built for versions
	Microsoft.EntityFrameworkCore.SqlServer [↗]	SQL Server 2012 onwards	EF Core Project [↗] (Microsoft)		6.0
	Microsoft.EntityFrameworkCore.Sqlite [↗]	SQLite 3.7 onwards	EF Core Project [↗] (Microsoft)		6.0
	Microsoft.EntityFrameworkCore.InMemory [↗]	EF Core in-memory database	EF Core Project [↗] (Microsoft)	Limitations	6.0
	Microsoft.EntityFrameworkCore.Cosmos [↗]	Azure Cosmos DB SQL API	EF Core Project [↗] (Microsoft)		6.0
	Npgsql.EntityFrameworkCore.PostgreSQL [↗]	PostgreSQL	Npgsql Development Team [↗]		6.0
	Pomelo.EntityFrameworkCore.MySql [↗]	MySQL, MariaDB	Pomelo Foundation Project [↗]		6.0
	MySql.EntityFrameworkCore [↗]	MySQL	MySQL project [↗] (Oracle)		6.0

EF Core Database Providers

